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ANSI/NETA ECS-2024

2024 ECS

STANDARD FOR

ELECTRICAL COMMISSIONING SPECIFICATIONS

FOR ELECTRICAL POWER EQUIPMENT & SYSTEMS

NETA®

ANSI/NETA ECS-2024

STANDARD FOR ELECTRICAL COMMISSIONING SPECIFICATIONS for Electrical Power Equipment and Systems

Secretariat
NETA (InterNational Electrical Testing Association)

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3. Qualifications
4. Division of Responsibility
5. General

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FOREWORD

(This Foreword is not part of American National Standard NETA ECS-2024)

The InterNational Electrical Testing Association (NETA) was formed in 1972 to establish uniform testing procedures for electrical equipment and systems. NETA has been an Accredited Standards Developer for the American National Standards Institute since 1996. NETA's scope of standards activity is different from that of IEEE, NECA, NEMA, and UL. In matters of testing electrical equipment and systems NETA continues to reference other standards developers' documents where applicable. NETA's review and updating of presently published standards takes into account both national and international standards. NETA's standards may be used internationally as well as in the United States. NETA firmly endorses a global standardization. IEC standards as well as American consensus standards are taken into consideration by NETA's ballot pools and reviewing committees.

The ANSI/NETA *Standard for Electrical Commissioning Specifications for Electrical Power Equipment and Systems* is the most current revision of this document and was approved as an American National Standard on July 2, 2024.

The ANSI/NETA *Standard for Electrical Commissioning Specifications for Electrical Power Equipment and Systems* should be used in conjunction with the most recent edition of ANSI/NETA *Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems*. Together, these standards provide the necessary specifications to test and commission electrical power equipment and systems. Furthermore, the NETA standards should be used together with other commissioning documents to expand the scope to include all of the applicable systems. Other systems may include mechanical, instrumentation, heating and refrigeration, and building systems.

The ANSI/NETA *Standard for Electrical Commissioning Specifications for Electrical Power Equipment and Systems* was developed for use by those responsible for testing and commissioning newly installed or retrofitted electrical power systems and equipment to guide them in specifying and performing the necessary inspections, tests, measurements, and system performance verifications to commission an electrical power system infrastructure. This document aids in ensuring safe, reliable operation of the electrical power equipment and systems. It is essential to commission newly installed and retrofitted electrical power equipment and systems. Additionally, acceptance testing of the equipment provides the baseline test results for maintenance programs and equipment trending, while commissioning verifies the electrical equipment and system meets the Owner's Project Requirements and Basis of Design.

PREFACE

(This Preface is not part of American National Standard NETA ECS-2024)

It is recognized by the Association that the term "commissioning" is not well defined in the industry. It is the intent of this document to better define and specifically address the critical elements and requirements necessary in the commissioning of low-, medium-, and high-voltage electrical power systems.

The NETA ECS *Standard for Electrical Commissioning Specifications for Electrical Power Equipment and Systems* is intended for use in all industries, whether it is a data center, a utility substation, an industrial facility, or any location that requires a comprehensive electrical power system commissioning process.

Suggestions for improvement of this standard are welcome. They should be sent to the InterNational Electrical Testing Association, 3050 Old Centre Road, Suite 101, Portage, MI 49024.

To help the user better understand and navigate more efficiently through this document, the following information is offered:

Notation of Changes
Material included in this edition of the document but not part of the previous edition is marked with a black vertical line in the outside margin.

Document Structure
The NETA ECS is divided into eleven separate and defined sections.

Section 1	General Scope
Section 2	Applicable References
Section 3	Qualifications of Commissioning Organization and Personnel
Section 4	Division of Responsibility
Section 5	General
Section 6	Commissioning Process
Section 7	General Inspection and Commissioning Procedures
Section 8	Source Specific Systems Commissioning
Section 9	Thermographic Survey
Section 10	Transfer to Owner/Operator
Appendices	Various Informational Documents

Section 7 Structure
Section 7 is the main body of the document with specific information on what to do relative to the inspection and commissioning of electrical power equipment and systems. This document does not explain how to test specific pieces of equipment or systems, but rather details the required steps and procedures for comprehensive commissioning of electrical power systems at a given voltage class or source specific systems.

Section 7 is divided into three subsections (7.1, Low-; 7.2, Medium-; and 7.3, High- and Extra-High Voltage Systems), which are in turn divided again into three subsections (A. Pre-Energization, B. Energization, and C. Post Energization). Source specific systems are included in section 8 and thermographic surveys are included in section 9.

If/When Applicable
The phrases "if applicable", "when applicable", and any variation thereof do not occur in this standard. This standard assumes that if devices or pieces of equipment are not present, they will not be subject to testing or verification.

Manufacturer's Instruction Manuals
It is important to follow the recommendations contained in the manufacturer's published data. Many of the details of a complete and effective testing and commissioning procedure can be obtained from this source.

Summary
The guidance of an experienced commissioning professional should be sought to develop a commissioning plan and lead the commissioning team. It is necessary to make an informed judgment for each particular system regarding how extensive a procedure is justified and required. The approach taken in these specifications is to present a comprehensive series of inspections, tests, and verifications applicable to most electrical systems. It is likely that in smaller electrical systems some of the inspections, tests, and verifications can be omitted.

Likewise, guidance of an experienced commissioning professional should also be sought when making decisions concerning the results of test data and performance results and their significance to the overall analysis of the device or system under test. Careful consideration of all aspects of test data, including manufacturer’s published data and recommendations, must be included in the overall assessment of the equipment or systems being commissioned.

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1. GENERAL SCOPE

1.1 Electrical Commissioning Specifications

1. These specifications describe the systematic process of documenting and placing into service newly–installed or retrofitted electrical power equipment and systems. This document shall be used in conjunction with the most recent edition of the ANSI/NETA *Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems* (ANSI/NETA ATS). The individual electrical components shall be subjected to factory and field tests, as required, to validate the individual components.

2. The purpose of these specifications is to assure that tested electrical systems are safe, reliable, operational, in conformance with applicable standards and manufacturers' tolerances, and installed in accordance with design specifications.

3. The work outlined in these specifications may involve hazardous voltages, materials, operations, and equipment. These specifications do not purport to address all of the safety concerns, if any, associated with their use. It is the responsibility of the user of this standard to establish appropriate safety, health, and environmental practices and determine the applicability of regulatory limitations prior to the use of this specification.

4. These specifications are intended for application on electrical power equipment and systems. It is not the intent of these specifications to provide comprehensive details on the commissioning of mechanical equipment, mechanical instrumentation systems, and related components.

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Specifications

All inspections and field tests shall be in accordance with the latest edition of the following codes, standards, and specifications, except as provided otherwise herein.

1. American Society of Heating, Refrigeration and Air-Conditioning Engineers – ASHRAE

ASHRAE Guideline 0	*The Commissioning Process*
ASHRAE Guideline 1.1	*HVAC&R Technical Requirements for the Commissioning Process*
ASHRAE 202	*Commissioning Process for Buildings and Systems*

2. Canadian Standards Association – CSA

CSA C22.1	*Canadian Electrical Code, Part 1, Safety Standard for Electrical Installations*
CSA Z462	*Workplace Electrical Safety Standard*
CSA Z463	*Maintenance Of Electrical Systems*

3. International Electrotechnical Commission - IEC

IEC 62446-1	*Photovoltaic (PV) Systems - Requirements for Testing, Documentation, and Maintenance - Part 1: Grid Connected Systems - Documentation, Commissioning Tests and Inspection*

4. InterNational Electrical Testing Association – NETA

ANSI/NETA ETT	*Standard for Certification of Electrical Testing Technicians*
ANSI/NETA ATS	*Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems*

5. National Electrical Contractors Association – NECA

NECA 90	*Standard for Commissioning Building Electrical Systems*

6. National Fire Protection Association – NFPA

NFPA 70	*National Electrical Code*
NFPA 70B	*Standard For Electrical Equipment Maintenance*
NFPA 70E	*Standard for Electrical Safety in the Workplace*

2. APPLICABLE REFERENCES

2.1 Codes, Standards, and Specifications *(continued)*

7. North American Electric Reliability Corporation – NERC

NERC MOD 025-2	*Verification and Data Reporting of Generator Real and Reactive Power Capability and Synchronous Condenser Reactive Power Capability*
NERC MOD 026-1	*Verification of Models and Data for Generator Excitation Control System or Plant Volt/Var Control Functions*
NERC MOD 027-1	*Verification of Models and Data for Turbine/Governor and Load Control or Active Power/Frequency Control Functions*

8. Occupational Safety and Health Administration – OSHA

OSHA 29 CFR Part 1910	*Occupational Safety and Health Standards*
OSHA 29 CFR Part 1926	*Safety and Health Regulations for Construction*

2. APPLICABLE REFERENCES

2.2 Other Publications

GSA Public Buildings Service, *Building Commissioning Guide*

2.3 Contact Information

American Society of Heating, Refrigeration, and Air-Conditioning Engineers – ASHRAE
ashrae.org

Canadian Standards Association – CSA
csa.ca

Federal Energy Regulatory Commission – FERC
ferc.gov

GSA Public Buildings Service
gsa.gov/pbs

International Electrotechnical Commission – IEC
iec.ch

Institute of Electrical and Electronics Engineers – IEEE
ieee.org

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National Electrical Contractors Association – NECA
necanet.org

National Fire Prevention Association – NFPA
NFPA.org

North American Electric Reliability Corporation – NERC
nerc.com

National Institute of Standards and Technology – NIST
nist.gov

Occupational Safety and Health Administration – OSHA
U.S. Department of Labor, Occupational Safety and Health Administration
osha.gov

3. QUALIFICATIONS OF COMMISSIONING ORGANIZATION AND PERSONNEL

3.1 Commissioning Organization

1. The commissioning organization shall be an independent, third-party entity which can function as an unbiased authority, professionally independent of the manufacturers, suppliers, and installers of equipment or systems being evaluated.

2. The commissioning organization shall be regularly engaged in the commissioning of electrical equipment, devices, installations, and systems.

3. The commissioning organization shall use personnel who are regularly employed for electrical commissioning services.

4. The commissioning organization shall submit appropriate documentation to demonstrate that it satisfactorily complies with these requirements.

3.2 Commissioning Personnel

Personnel performing these commissioning activities shall be trained and experienced concerning the apparatus and systems being evaluated. These individuals shall be capable of conducting the tests in a safe manner and with complete knowledge of the hazards involved. They must evaluate the test data and make a judgment on the serviceability of the specific equipment.

4. DIVISION OF RESPONSIBILITY

4.1 The Owner's Representative

The owner's representative shall provide the commissioning organization with the following:

1. A short-circuit analysis, a coordination study, an incident energy analysis, and a protective device setting sheet as described in ANSI/NETA ATS, Section 6.
2. Most recent version of the electronic setting files for intelligent electronic devices and relay logic diagrams.
3. A complete set of as-built electrical plans, product data submittals, and specifications.
4. Drawings and instruction manuals applicable to the scope of work.
5. Factory and field acceptance test reports.
6. Notification of when equipment becomes available for commissioning work. Work shall be coordinated to expedite project scheduling.
7. Project schedule.
8. Site-specific hazard notification and safety training.
9. Owner's Project Requirements (OPR).
10. Basis of Design (BOD) document.
11. Designated representative(s) for the project commissioning activities.

4.2 The Commissioning Organization

The commissioning organization shall provide the following:

1. All necessary services and technical expertise to conduct electrical commissioning.
2. Notification to the designated representative(s) prior to the commencement of any electrical commissioning activity.
3. Timely notification of deficiencies based on the results of the commissioning activities.
4. A written record of all electrical commissioning activities and a final report.

5. GENERAL

5.1 Safety and Precautions

All parties involved must be cognizant of industry-standard safety procedures. This document does not include any procedures, including specific safety procedures. It is recognized that an overwhelming majority of the activities recommended in these specifications are potentially hazardous. Individuals conducting these activities shall be capable of conducting them in a safe manner and with complete knowledge of the hazards involved.

1. Safety practices shall include, but are not limited to, the following requirements:

    1. All applicable provisions of the Occupational Safety and Health Act, particularly OSHA 29CFR 1910 and 29 CFR Part 1926.

    2. NFPA 70E, *Standard for Electrical Safety in the Workplace* and CSA Z462, *Workplace Electrical Safety*.

    3. Applicable state and local safety operating procedures.

    4. Owner's safety practices.

2. The commissioning organization shall have a designated safety lead person on site to supervise operations with respect to safety.

3. A job hazard analysis and a safety briefing shall be conducted prior to the commencement of work.

4. The commissioning organization shall have a designated safety representative assigned to the project to supervise operations with respect to safety. This individual may be the same person described in 5.1.2.

5.2 Suitability of Test Equipment

1. All test equipment shall meet the requirements in Section 5.3 and be in good mechanical and electrical condition.

2. Field test metering used to check power system meter calibration must be more accurate than the instrument being tested.

3. Accuracy of metering in test equipment shall be appropriate for the test being performed.

4. Waveshape and frequency of test equipment output waveforms shall be appropriate for the test and the tested equipment.

5. GENERAL

5.3 Test Instrument Calibration

1. The commissioning organization shall have a calibration program which assures that all applicable test instruments are maintained within rated accuracy for each test instrument calibrated.

2. The firm providing calibration service shall maintain up-to-date instrument calibration instructions and procedures for each test instrument calibrated.

3. The accuracy shall be directly traceable to the National Institute of Standards and Technology (NIST) or ISO/IEC 17025, *General Requirements for the Competence of Testing and Calibration Laboratories.*

4. All test instruments shall be calibrated within 12 months of the date of test.

5. Dated calibration labels shall be visible on all test equipment.

6. Records, which show date and results of instruments calibrated or tested, must be kept up-to-date.

7. Calibrating standard shall be of higher accuracy than that of the instrument tested.

5. GENERAL

5.4 Documentation

The commissioning organization shall furnish a copy or copies of the complete documentation as specified in the commissioning contract. The commissioning documentation shall include the following:

1. Report
    1. Summary of project.
    2. Description of electrical system.
    3. The final commissioning plan.
    4. A copy of the logs and records of document reviews performed.
    5. A complete copy of the acceptance testing and functional performance test results.
    6. Identification of systems or assemblies that do not meet the Owner's Project Requirements.
    7. Analysis and recommendations.
    8. Resolution plan for incomplete tasks.
    9. As-left intelligent electronic device (IED) settings, documentation, and files.
    10. Commissioning Progress Reports.
    11. Commissioning Issues Log listing all closed and open issues.
2. Drawing packages
    1. All applicable drawing packages shall be as-built.
    2. A complete as-built drawing package shall be left on site and a duplicate as-built package shall be submitted to the owner/operator.
3. Operations & Maintenance (O&M) Manuals

    Applicable equipment manuals and operational instructions shall be readily available for the owner and operator.

6. COMMISSIONING PROCESS

6.1 Commissioning Intent

1. Commissioning is the systematic process of verifying, documenting, and placing into service newly-installed or retrofitted electrical power equipment and systems.

2. Commissioning is critical for all new or retrofit installation projects to verify the correct system operation to the design intent, thus contributing to the safe and reliable operation of the system.

3. The commissioning process involves Owner’s Project Requirements (OPR), The Basis of Design (BOD), factory acceptance tests, field acceptance tests, verification of the component interconnections, and functional testing of the system in part and in whole.

4. Acceptance tests and commissioning work provides baseline results for routine maintenance of the system and related components.

6.2 Owner’s Project Requirements

OPR is a written document that details the functional requirements of a project and the expectations of how it will be used and operated. This includes project goals, measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information. The terms project intent or design intent are used by some owners for their commissioning process Owner’s Project Requirements. (ASHRAE Guideline 0)

6.3 Basis of Design

The Basis of Design (BOD) is a document that records the concepts, calculations, decisions, and product selections used to meet the OPR and to satisfy applicable regulatory requirements, standards, and guidelines. The document includes both narrative descriptions and lists of individual items that support the design process. (ASHRAE Guideline 0)

6.4 Commissioning Plan

1. A commissioning plan is a document that outlines the organization, schedule, allocation of resources, and documentation requirements of the commissioning process.

2. The commissioning plan should be developed by the commissioning authority during the design phase of the project. The commissioning plan should be updated and expanded during the construction, acceptance, and occupancy phases of the project by the commissioning team.

3. Parties involved in the execution of the commissioning plan shall work from the most up-to-date version.

4. The commissioning plan should include the following information:

    1. Overview of the commissioning phases and activities.

6. COMMISSIONING PROCESS

6.4. Commissioning Plan (*continued*)

2. Roles and responsibilities for the commissioning team throughout the project.

3. Documentation of the general communication channels and project hierarchy.

4. Detailed commissioning schedule of the project, including design, construction, acceptance, occupancy stages, and key commissioning milestones.

5. General description of commissioning activities that will occur during pre-energization, energization, and post-energization.

6. Equipment and systems being commissioned.

7. Commissioning checklists, test plans, and test forms specific to the project.

8. A process for approval by the owner and operator to allow the equipment to be energized.

9. A process for approval by the owner and turnover of the project and equipment from the commissioning team to the owner.

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7. GENERAL INSPECTION AND TEST PROCEDURES

7.1 Low-Voltage Systems ($\leq$ 1 kV)

A. Pre-Energization

Prior to complete system energization, verify the following pre-energization tasks are complete:

1. Review Owner's Project Requirements (OPR), Basis of Design (BOD), project specifications, and regulatory requirements for information specifically related to the commissioning of the electrical system.

2. Equipment manuals and operational instructions shall be readily available for the commissioning team and owner.

3. Verify all equipment has been tested according to the most recent edition of the ANSI/NETA *Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems* (ANSI/NETA ATS) and the OPR.

4. Review factory and field acceptance test data, documentation, results, and deficiencies to verify acceptable condition and suitability for initial energization and final acceptance.

5. Verify nameplate and equipment ratings are documented and correct in accordance with the most current drawings.

6. Review drawings, specifications, submittals, logic diagrams, protective device settings, engineering studies, and other pertinent information to verify completeness.

7. Visually inspect equipment.

8. Verify equipment is clearly labeled with unique designations and match designations on all drawings, documentation, programming, and communication protocols.

9. Verify equipment, doors, and fences are labeled with appropriate safety labeling and have the correct information in accordance with applicable industry standards and codes.

10. Verify equipment, structures, and circuits are correctly bonded and grounded in accordance with applicable industry standards and codes.

11. Verify equipment can be safely isolated. Confirm isolation points can be visually confirmed open and a lock and/or tag can be applied.

12. Confirm that correct electrical equipment clearances have been met.

13. Confirm clear working space around electrical equipment in accordance with applicable industry standards and codes.

14. Verify correct operation of mechanical, electrical, key, and safety interlocks on electrical power equipment. Verify duplicate interlock keys are destroyed or retained by authorized personnel in accordance with manufacturer's published data.

7. GENERAL INSPECTION AND TEST PROCEDURES

7.1 Low-Voltage Systems ($\leq$ 1 kV) (*continued*)

15. Verify correct operation and sequencing of limit switches.

16. Verify documentation of control and interconnection wiring including shipping splits, field-connected devices, internal wiring, and SCADA interfaces.

17. Verify current transformer circuits are complete, have a single point ground, and do not have an open circuit. Shorting devices should be in the intended position.

18. Verify instrument transformer tap connections are correct and match documentation, drawings, and protective device settings.

19. Verify protective device settings and logic match documentation, drawings, and engineering studies. Verify logging, Sequence of Events (SOE), waveform capture, triggers, etc. are correctly set up. Verify firmware is up to date.

20. Verify arc-flash hazard warning labels are in accordance with applicable industry standards, codes, and engineering studies.

21. Verify correct operation of arc energy reduction systems.

22. Verify the field marking of maximum available fault current at service entrance equipment in accordance with applicable industry standards and codes and engineering studies.

23. Verify intelligent electronic devices, communication protocols, and SCADA systems properly trigger events and disturbance records.

24. Verify intelligent electronic devices, communication protocols, and SCADA systems display the correct date and time.

25. Verify all communication points to end device(s).

26. Verify neutral grounding system(s) in accordance with the design.

27. Verify zone-selective interlocking systems, are wired, set, and test results are available.

28. Verify accuracy of panelboard schedules.

29. Verify pre-startup procedures on prime, emergency, and standby power systems have been performed and documented and systems are operational and ready for energization.

30. Verify operation of transfer schemes.

31. Verify correct settings of lighting controls for electrical rooms and substations.

32. Verify correct operation of emergency shutdown systems.

33. Verify current and voltage sensors were tested in accordance with manufacturer's published data.

34. Verify transformer insulating fluid analysis was completed and results are acceptable.

35. Verify correct liquid insulating fluid levels and gas pressures in transformers.

36. Verify gas and insulating fluid alarm levels on electrical equipment.

37. Verify valves on electrical equipment are in the correct position for energization.

38. Verify transformer taps are in the correct position(s) for energization.

39. Verify installation, setup, and operation of online condition monitoring equipment.

40. Verify pre-start-up procedures on all battery system components have been performed and documented. Confirm ventilation and gas alarms are functioning properly.

41. Confirm the initializing charge has been completed for battery systems in accordance with manufacturer's requirements.

42. Verify that indications and records for faults, alarms, and metering are cleared.

43. Compile as-left setting files for electrical system infrastructure.

44. Submit test data, as-built drawings, and relay as-left files to owner prior to energizing equipment.

45. Develop a written energization plan.

B. Energization

1. Restrict access to electrical equipment during energization and commissioning activities.

2. Verify emergency exits are correctly secured to prevent the entrance of an unauthorized person from outside, while allowing free escape from inside.

3. Verify removal of temporary protective grounding equipment

4. Verify correct position of switches, circuit breakers, and transfer switches for control circuits, instrument transformer circuits, and power circuits.

5. Verify test switches and terminal block disconnects/switches are in the correct position in accordance with energization plan.

6. Follow and document the steps of the energization plan.

7. Verify correct current and voltage values to protective devices and metering.

7. GENERAL INSPECTION AND TEST PROCEDURES

7.1 Low-Voltage Systems ($\leq$ 1 kV) (*continued*)

8. Verify correct system phase angle and sequence.

9. Verify equipment phasing.

10. Verify load test in accordance with the OPR.

11. Verify correct motor rotation.

12. Verify battery and uninterruptible power supply (UPS) systems are free of alarms and are in the specified operating mode.

13. Verify no alarms or fault indications are present.

14. Verify prime, emergency, and standby power systems are free of alarms and are in the specified operating mode.

C. Post Energization

1. Obtain post-energization insulating fluid and gas analysis on electrical equipment.

2. Verify monitoring and protective devices have established baseline parameters and criteria.

3. Perform thermographic survey of equipment in accordance with Section 9.

4. For warranty documentation, record initial energization date for transmittal to owner and manufacturer.

5. Verify equipment loading and compare to design criteria.

6. Verify balanced loading and voltage levels on all panelboards.

7. Verify power quality survey in accordance with the OPR.

8. Verify all equipment and systems functioned as per the OPR, BOD, and approved sequence of operations.

9. Develop commissioning report and documentation in accordance with Section 5.4.

10. Verify correct operation of GFCI receptacles.

7. GENERAL INSPECTION AND TEST PROCEDURES

7.2 Medium-Voltage Systems, (> 1 kV and < 100 kV)

A. Pre-Energization

1. Review the Owner's Project Requirements (OPR), Basis of Design (BOD), project specifications, and regulatory requirements for information specifically related to the commissioning of the electrical system.

2. Equipment manuals and operational instructions shall be readily available for the commissioning team and the owner.

3. Verify all equipment has been tested according to the most recent edition of the ANSI/NETA *Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems* (ANSI/NETA ATS) and the OPR.

4. Review factory and field acceptance test data, documentation, results, and deficiencies to verify acceptable condition and suitability for initial energization and final acceptance.

5. Verify nameplate and equipment ratings are documented and correct in accordance with the most current drawings.

6. Review drawings, specifications, submittals, logic diagrams, protective device settings, engineering studies, and other pertinent information to verify completeness.

7. Visually inspect equipment.

8. Verify equipment is clearly labeled with unique designations and match designations on drawings, documentation, programming, and communication protocols.

9. Verify equipment, doors, and fences are labeled with appropriate safety labeling and have the correct information in accordance with applicable industry standards and codes.

10. Verify equipment, structures, and circuits are correctly bonded and grounded in accordance with applicable industry standards and codes.

11. Verify medium-voltage components have adequate means for applying temporary protective grounding equipment.

12. Verify equipment can be safely isolated. Confirm isolation points can be visually confirmed open and a lock and/or tag can be applied.

13. Confirm that correct electrical equipment clearances have been met.

14. Verify arc-flash hazard warning labels are in accordance with applicable industry standards and codes and engineering studies.

7. GENERAL INSPECTION AND TEST PROCEDURES

7.2 Medium-Voltage Systems, (> 1 kV and < 100 kV) (*continued*)

15. Verify correct operation of arc energy reduction systems.

16. Verify mechanical, electrical, key, and safety interlocks on electrical power equipment. Verify duplicate interlock keys are destroyed or retained by authorized personnel in accordance with manufacturer's published data.

17. Verify correct operation and sequencing of limit switches.

18. Verify documentation of control and interconnection wiring including shipping splits, field-connected devices, internal wiring, and SCADA interface.

19. Verify current transformer circuits are complete, have a single point ground, and do not have an open circuit. Shorting devices should be in the intended position.

20. Verify instrument transformer tap connections are correct and match documentation, most current drawings, and protective device settings.

21. Verify protective device settings and logic match documentation, drawings, and engineering studies. Verify logging, Sequence of Events (SOE), waveform capture, triggers, etc. are correctly set up. Verify firmware is up to date.

22. Verify correct operation of applicable devices for protection and control schemes, SCADA, and communication protocols.

23. Verify intelligent electronic devices, communication protocols, and SCADA systems properly trigger events and disturbance records.

24. Verify intelligent electronic devices, communication protocols, and SCADA systems are connected to an adequate time synchronization source and all devices display the correct date and time.

25. Verify all communication points to end device(s).

26. Verify neutral grounding system(s) in accordance with the design.

27. Verify correct operation of communication assisted protection schemes.

28. Verify pre-startup procedures on prime, emergency, and standby power systems have been performed and documented and systems are operational and ready for energization.

29. Verify operation of transfer schemes.

30. Verify correct operation of lighting controls for electrical rooms and substations.

31. Verify correct operation of emergency shutdown systems.

32. Verify current and voltage sensors were tested in accordance with manufacturer's published data.

33. Verify differential schemes operate in accordance with OPR.

34. Perform system primary injection (through-fault) testing in accordance with the OPR.

35. Verify transformer and circuit breaker insulating fluid and gas analysis was completed and results are acceptable.

36. Verify correct insulating fluid levels and gas pressures in electrical equipment.

37. Verify proper spill containment and signage is present around all liquid filled transformers and similar devices where hazards exist.

38. Verify valves on electrical equipment are in the correct position for energization.

39. Verify transformer taps are in the correct position(s) for energization.

40. Verify installation, setup, and operation of online condition monitoring equipment.

41. Verify pre-start-up procedures on all battery system components have been performed and documented. Confirm ventilation and gas alarms are functioning properly.

42. Confirm the initializing charge has been completed for the battery systems and float voltage, equalize voltage, current limits, and alarms are in accordance with manufacturers' requirements. Confirm ventilation and gas alarms are functioning properly.

43. Verify that indications and records for faults, alarms, and metering are cleared.

44. Compile as-left setting files for electrical system infrastructure.

45. Submit test data, as-built drawings, and relay as-left files to owner prior to energizing equipment.

46. Develop a written energization plan.

B. Energization

1. Restrict access to electrical equipment during energization and commissioning activities.

2. Verify emergency exits are correctly secured to prevent the entrance of an unauthorized person from outside, while allowing free escape from inside.

3. Verify removal of temporary protective grounding equipment.

4. Verify correct position of switches, circuit breakers, and transfer switches for control circuits, instrument transformer circuits, and power circuits.

5. Verify test switches and terminal block disconnects/switches are in the correct position in accordance with the energization plan.

6. Follow and document the steps of the energization plan.

7. Verify correct operate and restraint values to differential protection.

8. Verify correct system phase angle and sequence.

9. Verify equipment phasing.

10. Verify correct motor rotation.

11. Verify transformer load tap changer and automatic voltage regulator operation.

12. Verify battery and UPS systems are free of alarms and are in the specified operating mode.

13. Verify monitoring devices are functioning correctly.

14. Verify temperature monitoring and protective devices have established baseline parameters and criteria.

15. Verify no alarms or fault indications are present.

16. Verify prime, emergency, and standby power systems are free of alarms and are in the specified operating mode.

C. Post Energization

1. Obtain post-energization insulating fluid and gas analysis on electrical equipment.

2. Verify monitoring and protective devices have established baseline parameters and criteria.

3. Verify any required regulatory compliance reports in accordance with the OPR.

4. For warranty documentation, record initial energization date for transmittal to owner and manufacturer.

5. Verify partial discharge survey of equipment in accordance with the OPR.

6. Perform thermographic survey of equipment in accordance with Section 9.

7. Verify equipment loading and compare to design criteria.

8. Verify correct operation of GFCI receptacles.

7.2 Medium-Voltage Systems, (> 1 kV and < 100 kV) (*continued*)

9. Verify correct current and voltage values to protective devices and metering.

10. Perform power quality survey in accordance with the OPR.

11. Verify all equipment and systems functioned as per the OPR, BOD, and approved sequence of operations.

12. Develop commissioning report and documentation in accordance with Section 5.4.

7. GENERAL INSPECTION AND TEST PROCEDURES

7.3 High-Voltage and Extra-High Voltage Systems ($\geq$ 100 kV and < 1,000 kV)

A. Pre-Energization

Prior to complete system energization, assure the following pre-energization tasks have been completed.

1. Review the Owner's Project Requirements (OPR), Basis of Design (BOD), project specifications, and regulatory requirements for information specifically related to the commissioning of the electrical system.

2. Equipment manuals and operational instructions shall be readily available for the commissioning team and the owner.

3. Verify all equipment has been tested according to the most recent edition of the ANSI/NETA *Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems* (ANSI/NETA ATS) and the OPR.

4. Review factory and field acceptance test data, documentation, results, and deficiencies to verify acceptable condition and suitability for initial energization and final acceptance.

5. Verify nameplate and equipment ratings are documented and correct in accordance with the most current drawings.

6. Review drawings, specifications, submittals, logic diagrams, protective device settings, engineering studies, and other pertinent information to verify completeness.

7. Visually inspect equipment.

8. Verify equipment is clearly labeled with unique designations and match designations on drawings, documentation, programming, and communication protocols.

9. Verify equipment, doors, and structures are labeled with appropriate safety labeling and have the correct information in accordance with applicable industry standards and codes.

10. Verify equipment and circuits are correctly bonded and grounded in accordance with applicable industry standards and codes.

11. Verify high-voltage components have adequate means for applying temporary protective grounding equipment.

12. Verify equipment can be safely isolated. Confirm isolation points can be visually confirmed open and a lock and/or tag can be applied.

13. Confirm that correct electrical equipment clearances have been met.

7. GENERAL INSPECTION AND TEST PROCEDURES

7.3 High-Voltage and Extra-High Voltage Systems ($\geq$ 100 kV and < 1,000 kV) (*continued*)

14. Verify correct operation of mechanical, electrical, and safety interlocks on electrical power equipment. Verify duplicate keys are destroyed or retained by authorized personnel in accordance with manufacturer's published data.

15. Verify correct operation and sequencing of limit switches.

16. Verify documentation of control and interconnection wiring including shipping splits, field-connected devices, internal wiring, and SCADA interfaces.

17. Verify current transformer circuits are complete, have a single point ground, and do not have an open circuit. Shorting devices should be in the intended position.

18. Verify instrument transformer tap connections are correct and match documentation, most current drawings, and protective device settings.

19. Verify protective device settings and logic match documentation, drawings, and engineering studies. Verify logging, Sequence of Events (SOE), waveform capture, triggers, etc. are correctly set up. Verify firmware is up to date.

20. Verify intelligent electronic devices, communication protocols, and SCADA systems properly trigger events and disturbance records.

21. Verify intelligent electronic devices, communication protocols, and SCADA systems are connected to an adequate time synchronization source and all devices display the correct date and time.

22. Verify all communication points to end device(s).

23. Verify correct operation of lighting controls for electrical rooms and substations.

24. Verify differential schemes operate in accordance with OPR.

25. Verify system primary injection (through-fault) testing in accordance with the OPR.

26. Verify correct operation of communication assisted protection schemes.

27. Verify current and voltage sensors were tested in accordance with manufacturer's published data.

28. Verify transformer and circuit breaker insulating fluid and gas analysis was completed and results are acceptable.

29. Verify correct insulating fluid levels and gas pressures in electrical equipment.

30. Verify valves on electrical equipment are in the correct position for energization.

7.3 High-Voltage and Extra-High Voltage Systems ($\geq$ 100 kV and < 1,000 kV) (*continued*)

31. Verify transformer taps are in the correct position(s) for energization.

32. Verify installation, setup, and operation of online condition monitoring equipment.

33. Verify pre-start-up procedures on all battery system components have been performed and documented. Confirm ventilation and gas alarms are functioning properly.

34. Confirm the initializing charge has been completed for the battery systems and float voltage, equalize voltage, current limits, and alarms are all in accordance with manufacturer's requirements. Confirm ventilation and gas alarms are functioning properly.

35. Verify that indications and records for faults, alarms, and metering are cleared.

36. Compile as-left setting files for electrical system infrastructure.

37. Submit test data, as-built drawings, and relay as-left files to owner prior to energizing equipment.

38. Develop a written energization plan.

B. Energization

1. Restrict access to the substation and components during energization and commissioning activities.

2. Verify removal of temporary protective grounding equipment.

3. Verify correct position of switches, circuit breakers, and transfer switches for control circuits, instrument transformer circuits, and power circuits.

4. Verify test switches and terminal block disconnects/switches are in the correct position in accordance with energization plan.

5. Follow and document the steps of the energization plan.

6. Verify correct current and voltage values to protective devices and metering.

7. Verify correct operate and restraint values on differential protection.

8. Verify correct system phase angles and sequence.

9. Verify equipment phasing.

10. Verify correct motor rotation for motors on electrical equipment and associated equipment.

11. Verify transformer load tap changer and automatic voltage regulator operation.

7. GENERAL INSPECTION AND TEST PROCEDURES

7.3 High-Voltage and Extra-High Voltage Systems ($\geq$ 100 kV and < 1,000 kV) (*continued*)

12. Verify battery and UPS systems are free of alarms and are in the specified operating mode.

13. Verify monitoring devices are functioning properly.

14. Verify temperature monitoring and protective devices have established baseline parameters and criteria.

15. Verify no alarms or fault indications are present.

C. Post-Energization

1. Obtain post-energization insulating fluid and gas analysis on electrical equipment.

2. Verify monitoring and protective devices have established baseline parameters and criteria.

3. Verify any required regulatory compliance reports in accordance with the OPR.

4. For warranty documentation, record initial energization date for transmittal to owner and manufacturer.

5. Verify partial discharge survey of equipment in accordance with the OPR.

6. Perform thermographic survey of equipment in accordance with Section 9.

7. Verify equipment loading and compare to design criteria.

8. Verify correct operation of GFCI receptacles.

9. Verify power quality survey in accordance with the OPR.

10. Verify all equipment and systems functioned as per the OPR, BOD, and approved sequence of operations.

11. Develop commissioning report and supply documentation in accordance with Section 5.4.

8. SOURCE SPECIFIC SYSTEMS COMMISSIONING

8.1 Solar Photovoltaic (PV) Commissioning Procedures

A. Pre-Energization

In addition to the Section 7 requirements for the applicable voltage classes (7.1, 7.2, and 7.3), verify or complete the following tasks specific to photovoltaic (PV) systems.

Prior to complete system energization, verify the following pre-energization tasks are complete:

1. Review Owner's Project Requirements (OPR), Basis of Design (BOD), project specifications, and regulatory requirements for information specifically related to the commissioning of the photovoltaic (PV) system.

2. Equipment manuals and operational instructions shall be readily available for the commissioning team and owner.

3. Verify all equipment has been tested according to the most recent edition of the ANSI/NETA Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems (ANSI/NETA ATS) and the OPR.

4. Review factory and field acceptance test data, documentation, results, and deficiencies to verify acceptable condition and suitability for initial energization and final acceptance.

5. Verify nameplate and equipment ratings are documented and correct in accordance with the most current drawings.

6. Review drawings, specifications, submittals, logic diagrams, protective device settings, engineering studies, and other pertinent information to verify completeness.

7. Visually inspect equipment.

8. Verify equipment is clearly labeled with unique designations and match designations on all drawings, documentation, programming, and communication protocols.

9. Verify equipment, doors, and fences are labeled with appropriate safety labeling and have the correct information in accordance with applicable industry standards and codes.

10. Verify equipment, structures, and circuits are correctly bonded and grounded in accordance with applicable industry standards and codes.

11. Verify equipment can be safely isolated. Confirm isolation points can be visually confirmed open and a lock and/or tag can be applied.

12. Confirm that correct electrical equipment clearances have been met.

13. Verify correct operation of mechanical, electrical, key, and safety interlocks on electrical power equipment. Verify duplicate interlock keys are destroyed or retained by authorized personnel in accordance with manufacturer's published data.

8.1 Solar Photovoltaic (PV) Commissioning Procedures *(continued)*

14. Verify correct operation and sequencing of limit switches.

15. Verify documentation of control and interconnection wiring including shipping splits, field-connected devices, internal wiring, and SCADA interface.

16. Verify current transformer circuits are complete, have a single point ground, do not have an open circuit. Shorting devices should be in the intended position.

17. Verify instrument transformer tap connections are correct and match documentation, drawings, and protective device settings.

18. Verify protective device settings and logic match documentation, drawings, engineering studies, and interconnection agreement. Verify logging, sequence of events (SOE), waveform capture, triggers, etc. are correctly set up. Verify firmware is up to date.

19. Verify arc-flash hazard warning labels are in accordance with applicable industry standards and codes and engineering studies.

20. Verify intelligent electronic devices, communication protocols, and SCADA systems correctly trigger events and disturbance records.

21. Verify intelligent electronic devices, communication protocols, and SCADA systems display the correct date and time.

22. Verify all communication points to end device(s).

23. Verify neutral grounding system(s) in accordance with the design.

24. Verify operation of transfer schemes.

25. Verify correct operation of emergency shutdown systems.

26. Verify current and voltage sensors were tested in accordance with manufacturer's published data.

27 Verify documentation for the grounding and bonding conductor continuity tests and the grounding system tests.

28. Verify documentation of the polarity tests for:

    a. Individual cables.

    b. PV string combiner boxes.

29. Verify documentation for blocking diode tests.

8.1 Solar Photovoltaic (PV) Commissioning Procedures *(continued)*

30. Verify documentation of the dry and wet insulation resistance tests for PV arrays or PV strings including modules, junction boxes, and cables.

31. Verify documentation of the acceptance tests for electrical equipment such as switches, circuit breakers, fuses, and surge protection devices.

32. Verify documentation for open-circuit voltage measurements for each PV string.

33. Verify documentation for short-circuit or operational current measurements for each PV string.

34. Complete the performance testing (I-V curve measurements).

35. Verify documentation for the setting, communication, data logging, and testing of the inverters.

36. Verify functional tests of the AC and DC system components including the inverters in accordance with manufacturer's published data.

37. Verify documentation for transfer trip testing.

38. Verify documentation for the setting, communication, data logging, and testing of phasor measurement units (PMUs) and phasor data concentrators (PDCs).

39. Verify that indications and records for faults, alarms, and metering are cleared.

40. Compile as-left setting files for electrical system infrastructure.

41. Submit test data, as-built drawings, and relay as-left files to owner prior to energizing equipment.

42. Develop a written energization plan.

8. SOURCE SPECIFIC SYSTEMS COMMISSIONING

8.1 Solar Photovoltaic (PV) Commissioning Procedures *(continued)*

B. Energization

1. Restrict access to electrical equipment during energization and commissioning activities.

2. Verify emergency exits are correctly secured to prevent the entrance of an unauthorized person from outside, while allowing free escape from inside.

3. Verify removal of temporary protective grounding equipment

4. Verify correct position of switches, circuit breakers, and transfer switches for control circuits, instrument transformer circuits, and power circuits.

5. Verify test switches and terminal block disconnects/switches are in the correct position in accordance with energization plan.

6. Follow and document the steps of the energization plan.

7. Verify correct current and voltage values to protective devices and metering.

8. Verify correct system phase angles and sequence.

9. Verify equipment phasing and synchronization at system connection points.

10. Verify documentation of the anti-islanding tests.

11. Verify system excitation, load control, grid code, and power system stabilizer testing and tuning.

12. Verify no alarms or fault indications are present.

8. SOURCE SPECIFIC SYSTEMS COMMISSIONING

8.1 Solar Photovoltaic (PV) Commissioning Procedures *(continued)*

C. Post-Energization

1. Verify monitoring and protective devices have established baseline parameters and criteria.

2. Perform thermographic survey of equipment in accordance with Section 9.

3. For warranty documentation, record initial energization date for transmittal to owner and manufacturer.

4. Verify equipment loading and compare to design criteria.

*5. Perform NERC MOD 025-2 reactive power capability verification.

*6. Perform NERC MOD 026-1 model verification for frequency response.

*7. Perform NERC MOD 027-1 model verification for voltage or reactive power response.

8. Verify phasor measurement system communications, readings, and data logging.

9. Verify transfer trip communications are healthy.

10. Verify SCADA communications are healthy.

11. Verify power quality survey in accordance with the OPR.

12. Verify all equipment and systems functioned as per the OPR, BOD, and approved sequence of operations.

13. Develop commissioning report and documentation in accordance with Section 5.4.

*Optional Test

8. SOURCE SPECIFIC SYSTEMS COMMISSIONING

8.2 Uninterruptible Power Supply (UPS) Commissioning Procedures

A. Pre-Energization

In addition to the Section 7 requirements for the applicable voltage classes (7.1, 7.2, and 7.3), verify or complete the following tasks specific to uninterruptible power supply (UPS) systems.

Prior to complete system energization, verify the following pre-energization tasks are complete:

1. Review Owner's Project Requirements (OPR), Basis of Design (BOD), project specifications, and regulatory requirements for information specifically related to the commissioning of the system.

2. Equipment manuals and operational instructions shall be readily available for the commissioning team and owner.

3. Verify all equipment has been tested according to the most recent edition of the ANSI/NETA Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems (ANSI/NETA ATS) and the OPR.

4. Review factory and field acceptance test data, documentation, results, and deficiencies to verify acceptable condition and suitability for initial energization and final acceptance.

5. Verify nameplate and equipment ratings are documented and correct in accordance with the most current drawings.

6. Review drawings, specifications, submittals, logic diagrams, protective device settings, engineering studies, and other pertinent information to verify completeness.

7. Visually inspect equipment.

8. Verify equipment is clearly labeled with unique designations and match designations on all drawings, documentation, programming, and communication protocols.

9. Verify equipment are labeled with appropriate safety labeling and have the correct information in accordance with applicable industry standards and codes.

10. Verify equipment, structures, and circuits are correctly bonded and grounded in accordance with applicable industry standards and codes.

11. Verify equipment can be safely isolated. Confirm isolation points can be visually confirmed open and a lock and/or tag can be applied.

12. Confirm that correct electrical equipment clearances have been met.

13. Confirm clear working space around electrical equipment in accordance with applicable industry standards and codes.

8.2 Uninterruptible Power Supply (UPS) Commissioning Procedures *(continued)*

14. Verify correct operation of mechanical, electrical, key, and safety interlocks on electrical power equipment. Verify duplicate interlock keys are destroyed or retained by authorized personnel in accordance with manufacturer's published data.

15. Verify correct operation and sequencing of limit switches.

16. Verify wiring interconnection points such as shipping splits, field-connected devices, and SCADA interface.

17. Verify current transformer circuits are complete, have a single point ground, and do not have an open circuit. Shorting devices should be in the intended position.

18. Verify instrument transformer tap connections are correct and match documentation, drawings, and protective device settings.

19. Verify protective device settings and logic match documentation, drawings, and engineering studies.

20. Verify arc-flash hazard warning labels are in accordance with applicable industry standards and codes and engineering studies.

21. Verify intelligent electronic devices, communication protocols, and SCADA systems properly trigger events and disturbance records.

22. Verify intelligent electronic devices, communication protocols, and SCADA systems display the correct date and time.

23. Verify all communication points to end device(s).

24. Verify neutral grounding system(s) in accordance with the design.

25. Verify operation of transfer schemes.

26. Verify correct operation of emergency shutdown systems.

27. Verify current and voltage sensors were tested in accordance with manufacturer's published data.

28. Verify pre-start-up procedures on all UPS and battery system components have been performed and documented.

29. Confirm the initializing charge has been completed for battery systems in accordance with manufacturer's requirements.

30. Verify that indications and records for faults, alarms, and metering are cleared.

31. Compile as-left setting files for electrical system infrastructure.

8. SOURCE SPECIFIC SYSTEMS COMMISSIONING

8.2 Uninterruptible Power Supply (UPS) Commissioning Procedures *(continued)*

32. Submit test data, as-built drawings, and relay as-left files to owner prior to energizing equipment.

33. Develop a written energization plan.

B. Energization

1. Restrict access to electrical equipment during energization and commissioning activities.

2. Verify emergency exits are correctly secured to prevent the entrance of an unauthorized person from outside, while allowing free escape from inside.

3. Verify removal of temporary protective grounding equipment

4. Verify correct position of switches, circuit breakers, and transfer switches for control circuits, instrument transformer circuits, and power circuits.

5. Verify test switches and terminal block disconnects/switches are in the correct position in accordance with energization plan.

6. Follow and document the steps of the energization plan.

7. Verify correct current and voltage values to protective devices and metering.

8. Verify correct system phase angle and sequence.

9. Verify equipment phasing.

10. Verify load test in accordance with the OPR.

11. Verify battery and UPS systems are free of alarms and are in the specified operating mode.

12. Verify no alarms or fault indications are present.

8. SOURCE SPECIFIC SYSTEMS COMMISSIONING

8.2 Uninterruptible Power Supply (UPS) Commissioning Procedures *(continued)*

C. Post Energization

1. Verify monitoring and protective devices have established baseline parameters and criteria.

2. Perform thermographic survey of equipment in accordance with Section 9.

3. For warranty documentation, record initial energization date for transmittal to owner and manufacturer.

4. Verify equipment loading and compare to design criteria.

5. Verify power quality survey in accordance with the OPR.

6. Verify all equipment and systems functioned as per the OPR, BOD, and approved sequence of operations.

7. Develop commissioning report and documentation in accordance with Section 5.4.

8. SOURCE SPECIFIC SYSTEMS COMMISSIONING

8.3 Automatic Transfer Switch (ATS) Commissioning Procedures

A. Pre-Energization

In addition to the Section 7 requirements for the applicable voltage classes (7.1, 7.2, and 7.3), verify or complete the following tasks specific to automatic transfer switch (ATS) systems.

Prior to complete system energization, verify the following pre-energization tasks are complete:

1. Review Owner’s Project Requirements (OPR), Basis of Design (BOD), project specifications, and regulatory requirements for information specifically related to the commissioning of the electrical system.

2. Read and understand the manufacturer's installation and operation manuals, as well as any specific documentation related to the ATS

3. Verify all equipment has been tested according to the most recent edition of the ANSI/NETA Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems (ANSI/NETA ATS) and the OPR.

4. Review factory and field acceptance test data, documentation, results, and deficiencies to verify acceptable condition and suitability for initial energization and final acceptance.

5. Verify the ATS is compatible with the electrical system and sources.

6. Verify nameplate and equipment ratings are documented and correct in accordance with the most current drawings.

7. Review drawings, specifications, submittals, logic diagrams, protective device settings, engineering studies, and other pertinent information to verify completeness.

8. Verify wiring connections and identification for the ATS sources and load.

9. Verify equipment is clearly labeled with unique designations and match designations on all drawings, documentation, programming, and communication protocols.

10. Verify equipment, doors, and fences are labeled with appropriate safety labeling and have the correct information in accordance with applicable industry standards and codes.

11. Verify equipment, structures, and circuits are correctly bonded and grounded in accordance with applicable industry standards and codes.

8. SOURCE SPECIFIC SYSTEMS COMMISSIONING

8.3 Automatic Transfer Switch (ATS) Commissioning Procedures *(continued)*

12. Verify equipment can be safely isolated. Confirm isolation points can be visually confirmed open and a lock and/or tag can be applied.

13. Confirm that correct electrical equipment clearances have been met.

14. Confirm clear working space around electrical equipment in accordance with applicable industry standards and codes.

15. Verify correct operation of mechanical, electrical, key, and safety interlocks on electrical power equipment. Verify duplicate interlock keys are destroyed or retained by authorized personnel in accordance with manufacturer's published data.

16. Verify correct operation and sequencing of limit switches.

17. Verify wiring interconnection points such as shipping splits, field-connected devices, and SCADA interface.

18. Verify current transformer circuits are complete, have a single point ground, and do not have an open circuit. Shorting devices should be in the intended position.

19. Verify instrument transformer tap connections are correct and match documentation, drawings, and protective device settings.

20. Verify protective device settings and logic match documentation, drawings, and engineering studies.

21. Verify arc-flash hazard warning labels are in accordance with applicable industry standards and codes and engineering studies.

22. Verify intelligent electronic devices, communication protocols, and SCADA systems properly trigger events and disturbance records.

23. Verify intelligent electronic devices, communication protocols, and SCADA systems display the correct date and time.

24. Verify all communication points to end device(s).

25. Verify neutral grounding system(s) in accordance with the design.

26. Verify operation of transfer schemes.

27. Verify current and voltage sensors were tested in accordance with manufacturer's published data.

28. Verify that indications and records for faults, alarms, and metering are cleared.

8.3 Automatic Transfer Switch (ATS) Commissioning Procedures *(continued)*

29. Test control circuitry of the ATS by simulating different power source scenarios Verify that the ATS responds correctly and automatically transfers power.

30. Develop a written energization plan.

B. Energization

1. Restrict access to electrical equipment during energization and commissioning activities.

2. Verify emergency exits are correctly secured to prevent the entrance of an unauthorized person from outside, while allowing free escape from inside.

3. Verify removal of temporary protective grounding equipment

4. Verify correct position of switches, circuit breakers, and transfer switches for control circuits, instrument transformer circuits, and power circuits.

5. Verify test switches and terminal block disconnects/switches are in the correct position in accordance with energization plan.

6. Follow and document the steps of the energization plan.

7. Verify correct current and voltage values to protective devices and metering.

8. Verify correct system phase angle and sequence.

9. Verify equipment phasing.

10. Conduct load transfer tests to verify the ATS transfers the connected load. Monitor the transfer time and verify it meets specified project requirements.

11. If the electrical system has multiple ATS units or is interconnected with other protective devices, conduct coordination tests to verify the proper sequencing and coordination of the ATS with other protective devices.

12. Check all alarms, indicators, and status lights on the ATS to ensure they are functioning correctly and provide accurate information about the power source status and ATS operation.

13. If the ATS supports remote monitoring and control capabilities, test these features to ensure proper communication, data exchange, and control functions.

C. Post Energization

1. Verify monitoring and protective devices have established baseline parameters and criteria.

8.3 Automatic Transfer Switch (ATS) Commissioning Procedures *(continued)*

2. Document all commissioning activities, including test results, adjustments made, and any issues encountered during the process.

3. Verify training to the owner-designees on the proper operation, maintenance, and troubleshooting of the ATS in accordance with the OPR.

4. Perform thermographic survey of equipment in accordance with Section 9.

5. For warranty documentation, record initial energization date for transmittal to owner and manufacturer.

6. Verify equipment loading and compare to design criteria.

7. Verify all equipment and systems functioned as per the OPR, BOD, and approved sequence of operations.

8. Develop commissioning report and documentation in accordance with Section 5.4.

9. THERMOGRAPHIC SURVEY

9.1 Visual and Mechanical Inspection

1. Remove panel covers or view the equipment through viewing ports designed to transmit applicable signals being measured.

*2. Perform a follow-up thermographic survey within 12 months of final acceptance by the owner.

9.2 Report

Provide a report which contains the following:

1. Description of equipment tested.
2. Discrepancies.
3. Temperature difference between the area of concern and the reference area.
4. Probable cause of temperature difference.
5. Areas inspected. Identify inaccessible and/or unobservable areas and/or equipment.
6. Load conditions at time of inspection.
7. Photographs and thermograms of the deficient area.
8. Recommended action.

9.3 Test Parameters

1. Inspect distribution systems with imaging equipment capable of detecting a minimum temperature difference of 1° C at 30° C.
2. Equipment shall detect emitted radiation and convert detected radiation to visual signal.
3. Thermographic surveys should be performed during periods of maximum possible loading, but not less than 40 percent of nominal circuit loading. Refer to NFPA 70B.

9.4 Test Results

For suggested actions based on temperature rise, refer to ANSI/NETA ATS, Table 100.18.

*Optional Test

10. Transfer to Owner/Operator

1. A formal and documented turnover of the project and the facility shall be performed.

2. The operator receiving the turnover shall be informed of any outstanding deficiencies and any abnormal operating conditions.

3. The turnover shall include all documentation, drawings, and operational responsibilities required.

4. Remove commissioning organization locks and tags and place operator locks and tags where required.

APPENDIX A
Definitions

NETA recognizes the *IEEE Standards Dictionary* as its official source for electrical definitions. The definitions in the list provided by NETA either are not included in the IEEE reference or are more specific to electrical testing and commissioning and to this document.

NETA defines equipment voltage ratings in accordance with ANSI/NEMA C37.84.1 *American National Standard for Electrical Power Systems and Equipment – Voltage Ratings (60 Hertz).*

As-found
Condition of the equipment prior to commissioning.

As-left
Condition of equipment at the completion of commissioning. As-left values refer to test values obtained after all commissioning tasks have been performed on the device or system under test.

Basis of Design (BOD)
A document that records the concepts, calculations, decisions, and product selections used to meet the Owner's Project Requirements (OPR) and to satisfy applicable regulatory requirements, standards, and guidelines. The document includes both narrative descriptions and lists of individual items that support the design process.

Comment
Suggested revision, addition, or deletion in an existing section of the NETA standards.

Commissioning
The systematic process of documenting and placing into service newly-installed or retrofitted electrical power equipment and systems.

Commissioning Authority
An entity identified by the Owner who leads, plans, schedules, and coordinates the commissioning team to implement the commissioning process.

Commissioning Plan
A document that outlines the organization, schedule, allocation of resources, and documentation requirements of the commissioning process.

Commissioning Process
A quality-focused process for enhancing the delivery of a project. The process focuses upon verifying and documenting that the facility and all of its systems and assemblies are planned, designed, installed, tested, operated, and maintained to meet the Owner’s Project Requirements.

Commissioning Team
Individuals who, through coordinated actions, are responsible for implementing the commissioning process.

APPENDIX A

Definitions *(continued)*

Electrical commissioning
See definition for *commissioning.*

Electrical tests
Electrical tests involve application of electrical signals and observation of the response. It may be, for instance, applying a potential across an insulation system and measuring the resultant leakage current magnitude or power factor/dissipation factor. It may also involve application of voltage and/or current to metering and relaying equipment to check for correct response.

Equipment condition
Suitability of the equipment for continued operation in the intended environment as determined by evaluation of the results of inspections and tests.

Exercise
To operate equipment in such a manner that it performs all its intended functions to allow observation, testing, measurement, and diagnosis of its operational condition.

Inspection
Examination or measurement to verify whether an item or activity conforms to specified requirements.

Interim amendment
An interim amendment is made by NETA’s Standards Review Council when a potential hazard is identified with respect to the use of this document. The interim amendment is in force from the time of its issue until the next revision of the document.

Manufacturer’s published data
Data provided by the manufacturer concerning a specific piece of equipment.

Mechanical inspection
Observation of the mechanical operation of equipment not requiring electrical stimulation, such as manual operation of circuit breaker trip and close functions. It may also include tightening of hardware, cleaning, and lubricating.

Owners Project Requirements (OPR)
A written document that details the functional requirements of a project and the expectations of how it will be used and operated. These include project goals, measurable performance criteria, cost considerations, benchmarks, success criteria, and supporting information. (The term project intent is used by some owners for their commissioning process Owner’s Project Requirements.)

The Owner's Representative
The Owner's Representative for the context of this document can be, but not be limited to, a Construction Manager, General Contractor, and/or Electrical Contractor that is contracted indirectly or directly with the Owner.

Re-Commissioning
An application of the commissioning process requirements to a project that has been delivered using the

commissioning process. This may be a scheduled re-commissioning developed as part of an ongoing commissioning process or it may be triggered by use change, operations problems, or other needs.

Ready-to-test condition
Having the equipment which is to be tested isolated, source and load disconnected, and temporary protective grounds applied to the system for safety as required.

Shall
Indicates a mandatory requirement and is used when the testing firm has control over the result.

Should
Indicates that among several possibilities one is recommended as particularly suitable, without mentioning or excluding others; or that a certain course of action is preferred but not necessarily required (*should* equals *is recommended that*).

System voltage
The root-mean-square (rms) phase-to-phase voltage of a portion of an alternating-current electric system. Each system voltage pertains to a portion of the system that is bounded by transformers or utilization equipment.

Verify
To investigate by observation or by test to determine that a particular condition exists.

Visual inspection
Qualitative observation of physical characteristics, including cleanliness, physical integrity, evidence of overheating, lubrication, etc.

Voltage Class:

NETA defines equipment voltage ratings in accordance with NEMA C37.84.1 American National Standard for Electrical Power Systems and Equipment – Voltage Ratings (60 Hertz).

Low voltage
A class of nominal system voltages 1000 volts or less.

Medium voltage
A class of nominal system voltages greater than 1000 volts and less than 100,000 volts.

High voltage
A class of nominal system voltages equal to or greater than 100,000 volts and equal to or less than 230,000 volts.

Extra-high voltage
A class of nominal system voltages greater than 230,000 volts.

APPENDIX B

Acronyms

AC	Alternating Current
ANSI	American National Standards Institute
ATS	ANSI/NETA ATS *Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems* (ATS can also mean Automatic Transfer Switch).
BOD	Basis of Design
CT	Current Transformer
CCVT	Capacitance-Coupled Voltage Transformer or CVT – capacitor voltage transformer
DC	Direct Current
DETC	Deenergized Tap Changer
ETT	ANSI/NETA ETT *Standard for Certification of Electrical Testing Technicians*
GFCI	Ground Fault Circuit Interrupter
HMI	Human-Machine Interface
IEC	International Electrotechnical Commission
IED	Intelligent Electronic Device
IEEE	Institute of Electrical and Electronics Engineers
IR	Infrared
LTC	Load Tap Changer
MTS	ANSI/NETA MTS *Standard for Maintenance Testing Specifications for Electrical Power Equipment and Systems*
NEC	National Electrical Code (NFPA 70)
NERC	North American Electric Reliability Corporation
NETA	InterNational Electrical Testing Association
NFPA	National Fire Protection Association
NIST	National Institute of Standards and Technology

APPENDIX B

Acronyms *(continued)*

O&M	Operations and Maintenance
OPR	Owner’s Project Requirements
OSHA	Occupational Safety and Health Administration
PT	Potential Transformer
RTU	Remote Terminal Unit
SCADA	Supervisory Control and Data Acquisition
SRC	Standards Review Council (NETA)
UPS	Uninterruptible Power Supply
ZSI	Zone Selective Interlocking

APPENDIX C

About the InterNational Electrical Testing Association

(The information contained in this appendix is not part of American National Standard (ANS) and has not been processed in accordance with ANSI's requirements for an ANS. As such, this appendix may contain material that has not been subjected to public review or a consensus process. In addition, it does not contain requirements necessary for conformance to the standard.)

The InterNational Electrical Testing Association (NETA) is an ANSI-accredited standards developer. NETA defines the standards by which electrical equipment is deemed safe and reliable. NETA Certified Technicians conduct the tests that ensure this equipment meets the Association's stringent specifications. NETA is the leading source of specifications, procedures, testing, and requirements, not only for commissioning new equipment but for testing the reliability and performance of existing equipment.

CERTIFICATION

Certification of competency is particularly important in the electrical testing industry. Inherent in the determination of the equipment's serviceability is the prerequisite that individuals performing the tests be capable of conducting the tests in a safe manner and with complete knowledge of the hazards involved. They must also evaluate the test data and make an informed judgment on the continued serviceability, deterioration, or nonserviceability of the specific equipment. NETA, a nationally-recognized certification agency, provides recognition of four levels of competency within the electrical testing industry in accordance with *ANSI/NETA ETT Standard for Certification of Electrical Testing Technicians.*

QUALIFICATIONS OF THE COMMISSIONING ORGANIZATION

An independent overview is the only method of determining the long-term usage of electrical apparatus and its suitability for the intended purpose. NETA Accredited Companies best support the interest of the owner, as the objectivity and competency of the commissioning firm is as important as the competency of the individual technician. NETA Accredited Companies are part of an independent, third-party electrical testing association dedicated to setting world standards in electrical maintenance and acceptance testing. Hiring a NETA Accredited Company assures the customer that:

- The NETA Technician has broad-based knowledge -- this person is trained to inspect, test, maintain, and calibrate all types of electrical equipment in all types of industries.
- NETA Technicians meet stringent educational and experience requirements in accordance with *ANSI/NETA ETT Standard for Certification of Electrical Testing Technicians*.
- A Registered Professional Engineer will review all engineering reports.
- All tests will be performed objectively, according to NETA specifications, using calibrated instruments traceable to the National Institute of Science and Technology (NIST).
- The firm is a well-established, full-service electrical testing business.

APPENDIX C

About the InterNational Electrical Testing Association
(continued)

SPECIFICATIONS AND PUBLICATIONS

As a part of its service to the industry, the InterNational Electrical Testing Association provides nationally-recognized publications:

ANSI/NETA ETT	*Standard for Certification of Electrical Testing Technicians*
ANSI/NETA MTS	*Standard for Maintenance Testing Specifications for Electrical Power Equipment and Systems*
ANSI/NETA ATS	*Standard for Acceptance Testing Specifications for Electrical Power Equipment and Systems*
ANSI/NETA ECS	*Standard for Electrical Commissioning Specifications for Electrical Power Equipment and Systems*

The Association also produces a quarterly technical journal, *NETA World*, which features articles of interest to electrical testing and maintenance companies, consultants, engineers, architects, and plant personnel directly involved in electrical testing and maintenance.

EDUCATIONAL PROGRAMS

PowerTest, NETA's annual technical conference, draws hundreds of qualified industry professionals from around the globe. This conference provides a forum for current industry advances, critical informational updates, networking, and more. Regular attendees include technicians from electrical testing and maintenance companies, consultants, engineers, architects, and plant personnel directly involved in electrical testing and maintenance. Technical presentations from field-experienced industry experts share practical knowledge and experience while in-depth seminars offer interactive training. At the Trade Show attendees enjoy the highest-quality gathering of industry-specific suppliers displaying state-of-the-art products and services directly related to the electrical testing industry. Attendance of PowerTest is the best opportunity for interaction and input in a professional technical environment.

APPENDIX D

Form for Comments

(The information contained in this appendix is not part of American National Standard (ANS) and has not been processed in accordance with ANSI's requirements for an ANS. As such, this appendix may contain material that has not been subjected to public review or a consensus process. In addition, it does not contain requirements necessary for conformance to the standard.)

Anyone may comment on this document using this form:

Type of Comment (Check one) ❑ Technical ❑ Editorial
Paragraph Number ______________________________
Recommend (Check One) ❑ *New Text ❑ *Revised Text ❑ *Deleted Text
❑This Comment is original material (Note: Original material is considered to be the submitter's own idea based on or as a result of his/her own experience, thought, or research and to the best of his/her knowledge is not copied from another source.)
❑This Comment is not original material; its source (if known) is ______________________________
Please Check One: ❑User ❑Producer ❑General Interest
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*1. All comments must be relevant to the proposed standard.
*2. Suggested changes must include (1) proposed text, including the wording to be added, revised (and how revised), or deleted, (2) a statement of the problem and substantiation for a technical change, and (3) signature of submitter. (Note: State the problem that will be resolved by your recommendation; give the specific reason for your comment, including copies of texts, research papers, testing experience, etc. If more than 200 words, it may be abstracted for publication.)
*3. Editorial comments are welcome, but they cannot serve as the sole basis for a suggested change.

A comment that does not include all required information may be rejected by the Standards Review Council for that reason. Must use separate form for each comment. All comments must be typed or printed neatly. Illegible comments will be interpreted to the best of the staff's ability.

This form is available electronically on NETA's website at www.netaworld.org/standards/standards-get-involved.

Send to: Standards Review Council
3050 Old Centre Road, Suite 101, Portage, MI 49024
Phone: 888.300.6382 FAX: 269.488.6383 Email: neta@netaworld.org

APPENDIX E

Form for Proposals

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Anyone may propose a new section for this document using the following form:

When drafting a proposed section:
Use the most recent edition of the specifications as a guideline for format and wording.
Remember that NETA specifications are "what to do" documents and do not include "how to do" information.
Include references.
When applicable, use the standard base format:

1. Pre-Energization
2. Energization
3. Post Energization

Date ____

Name ______________________________ Tel No. ______________

Company ___________________________ Fax No. ______________

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Please indicate organization represented (if any) ______________________

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Section/Number ______________________________________

Note 1: Type or print legibly in black ink.
Note 2: If supplementary material (photographs, diagrams, reports, etc.) is included, you may be required to submit additional copies.

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